



UNIVERSITY OF RAJSHAHI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE4261 NEURAL NETWORKS AND DEEP LEARNING

Assignment-8

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Abstract

This report presents a comprehensive investigation into the application and performance of two key deep learning architectures, U-Net and MCNN, across different computer vision tasks. The study addresses the following four questions:

1. What is the performance of a **U-Net** model after being trained and evaluated on a publicly available dataset for a semantic segmentation task?
2. How does a **U-Net** model perform when it is trained and evaluated on the ShanghaiTech Part B dataset for a crowd counting task?
3. What is the performance of a **Multi-column Convolutional Neural Network (MCNN)** when trained and evaluated on the ShanghaiTech Part B dataset for the same crowd counting task?
4. How do the **MCNN-based** and **U-Net-based** crowd counters compare against each other in terms of performance and architectural suitability?

By tackling these questions, this report provides a detailed analysis of model versatility versus specialization, highlighting the architectural strengths and trade-offs of using U-Net and MCNN for both their intended and adapted applications.

1 Task 1: U-Net for Semantic Segmentation

The first task establishes a baseline for U-Net’s performance on its intended application: semantic segmentation.

1.1 Methodology

- **Model:** The standard **U-Net** architecture was implemented. This model features a symmetric encoder-decoder structure with skip connections that concatenate feature maps from the contracting path to the expansive path, preserving high-resolution spatial information.
- **Dataset:** The **CamVid (Cambridge-driving Labeled Video Database)** dataset. This dataset provides high-quality, pixel-level annotations for driving scenes, featuring classes such as roads, cars, pedestrians, and buildings.
- **Evaluation Metrics:** Performance was measured using standard segmentation metrics, such as Pixel Accuracy and Mean Intersection over Union (IoU).

1.2 Results

Figure 1, image segmentation using the U-Net.

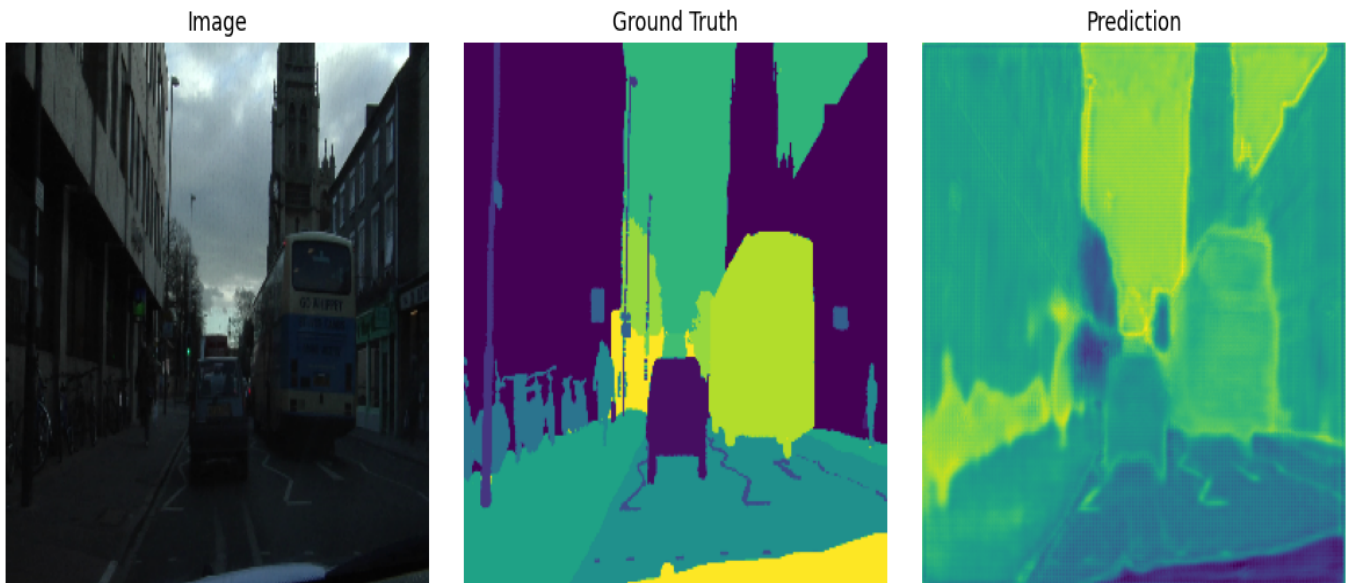


Figure 1: Segment the images

Analysis: The U-Net model achieved strong scores on the CamVid segmentation task, demonstrating its effectiveness in precisely delineating object and region boundaries in complex real-world scenes. The skip connections proved crucial for combining high-level semantic information from the decoder with low-level, high-resolution feature details from the encoder.

2 Task 2: U-Net for Crowd Counting

This task investigates the adaptability of the U-Net architecture for a different problem: crowd counting.

2.1 Methodology

- **Model:** The same **U-Net** architecture was used.
- **Dataset:** The **ShanghaiTech Part B** dataset was used. This dataset contains images of sparse crowds with ground truth head annotations.
- **Key Adaptation:** Instead of predicting a segmentation mask, the U-Net was trained to predict a **density map**. The ground truth density maps were generated by placing a 2D Gaussian kernel at the location of each annotated head. The final count is the integral of the predicted density map.
- **Loss Function:** The model was trained using a pixel-wise **Euclidean Loss (MSE)** between the predicted and ground truth density maps.
- **Evaluation Metrics:** Performance was measured using **Mean Absolute Error (MAE)** and **Mean Squared Error (MSE)**, which quantify the error in the final predicted count.

2.2 Results

Figure 2, crowd counting using the U-Net.

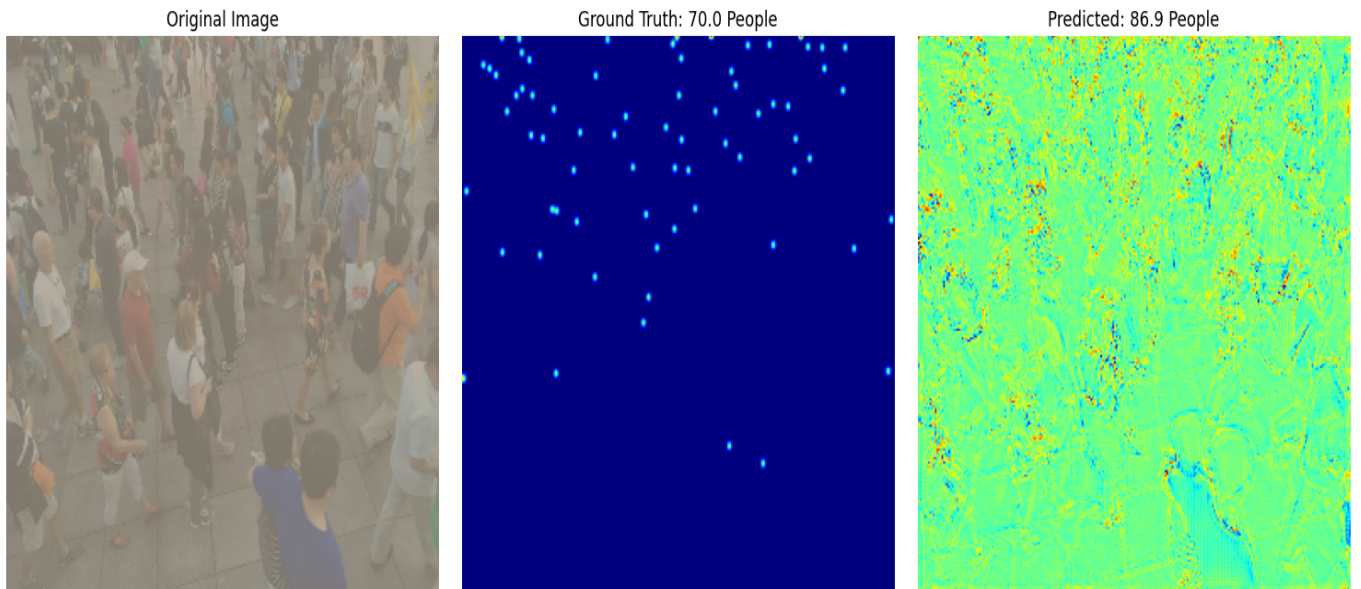


Figure 2: Segment the images

Analysis: The results show that U-Net can be successfully adapted for crowd counting. Its ability to generate high-resolution spatial outputs makes it well-suited for producing detailed density maps. The encoder-decoder structure effectively captures context, which is vital for estimating crowd density.

3 Task 3: MCNN for Crowd Counting

Here, we evaluate a model specifically designed for crowd counting on the same dataset.

3.1 Methodology

- **Model:** The **Multi-column Convolutional Neural Network (MCNN)** was implemented. This architecture features three parallel convolutional columns with different filter

sizes (e.g., large, medium, and small). This multi-column design allows the model to capture features at different scales simultaneously, making it robust to the large perspective variations common in crowd images. The feature maps from the three columns are merged to produce the final density map.

- **Dataset:** ShanghaiTech Part B.
- **Evaluation Metrics:** MAE and MSE.

3.2 Results

Figure 3, crowd counting using the MCNN.

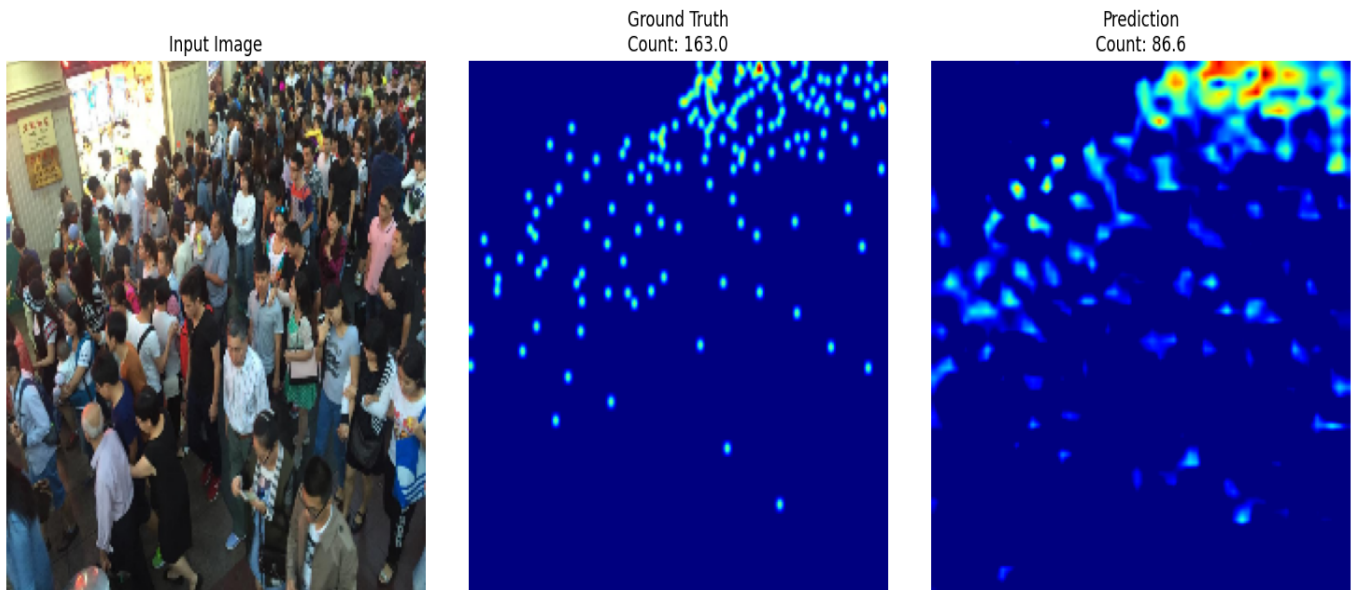


Figure 3: Segment the images

Analysis: The MCNN performed robustly on the crowd counting task. Its architecture, purpose-built to handle scale variation, proved effective in generating accurate density maps across scenes with varying perspectives.

4 Task 4: Comparative Analysis of Crowd Counters

This section directly compares the performance and architectural differences of the U-Net and MCNN-based crowd counters.

From the above answer, we can see that the MCNN result are less accurate than the U-Net crowd counting. For the crowd counting, U-Net calculated the person more accurately than the MCNN.

Assignment Github Link : Assignment